INTERNSHIP REPORT

1. Introduction

This internship offered practical exposure to **Data Analytics** through **Microsoft Power BI**, allowing me to work on real-world business scenarios. The primary objective was to analyse a Twitter dataset and develop interactive dashboards that provide insights into tweet performance, audience engagement, and user interaction.

The dataset contained various attributes, including Tweet ID, Date, Time, Impressions, Engagements, Engagement Rate, Likes, Retweets, Replies, URL Clicks, Hashtag Clicks, User Profile Clicks, Media Views, Media Engagements, App Opens, App Installs, Follows, and several other engagement metrics.

Throughout the internship, I converted raw data into meaningful visual reports using Power BI, which enhanced my analytical thinking, reporting abilities, and technical expertise.

2. Background

With the rapid growth of social media, organizations rely on data analytics to evaluate audience behaviour and content effectiveness. This project focused on analysing Twitter data in Microsoft Power BI by applying business rules and creating insightful dashboards.

The project involved:

- Cleaning and transforming data.
- Creating calculated columns and DAX measures.
- Applying conditional and time-based filters.
- Designing interactive dashboards.
- Performing performance and trend analysis.

3. Learning Objectives

During this internship, I gained experience in:

- Working efficiently with Microsoft Power BI.
- Performing data preparation and transformation.
- Writing DAX measures and calculated columns.
- Developing interactive and dynamic dashboards.
- Evaluating social media engagement metrics.
- Understanding ETL (Extract, Transform, Load) concepts.
- Strengthening analytical and logical thinking skills.

4. Activities and Tasks Performed

Task 1: Tweet Interaction Analysis by Category

Developed a clustered bar chart to compare URL Clicks, User Profile Clicks, and Hashtag Clicks across different tweet categories.

Business Rules Applied

- Dashboard visible only between 3:00 PM and 5:00 PM IST.
- Even-numbered tweet dates.
- Tweet word count greater than 40.
- At least one recorded interaction.

Result

After applying all the required business conditions, **no records satisfied the filtering criteria**, resulting in a blank visualization. This indicates that the specified conditions were successfully applied, and the absence of data reflects the dataset limitations rather than an implementation issue.

Task 2: Engagement Rate Comparison

Created a comparison of average engagement rates between tweets that generated App Opens and those that did not.

Business Rules Applied

- Tweets posted between 9:00 AM and 5:00 PM.
- Weekdays only.
- Even-numbered impressions.
- Odd-numbered dates.
- Character count above 30.
- Excluded tweets containing the letter "D".
- Dashboard visible only from 7:00–11:00 AM and 12:00–6:00 PM IST.

Result

Only the **"App Closed"** category satisfied all the specified conditions. The average engagement rate displayed is approximately **0.04**, showing that no tweets with App Opens met every filtering requirement.

Task 3: Media Interaction by Weekday

Designed a dual-axis chart comparing Media Views with Media Engagements across the days of the week.

Business Rules Applied

- Quarter 4 data only.
- Even-numbered impressions.
- Odd-numbered dates.
- Character count greater than 30.
- Excluded tweets containing the letter "H".
- Visible only between 7:00–11:00 AM and 3:00–5:00 PM IST.

Result

The dashboard shows **89K Media Views** and **62K Media Engagements** after applying all business filters. These values indicate that media content generated significant audience interaction within the filtered dataset

Task 4: Comparison of Replies, Retweets, and Likes

Created a comparison chart displaying the total number of Replies, Retweets, and Likes for tweets posted from June through August 2020.

Result

Among the three months, **July** recorded the highest overall engagement. **Likes** contributed the largest share of interactions, followed by Retweets and Replies, indicating that July had the most successful tweet performance.

Task 5: Monthly Engagement Rate Trend

Built a line chart to compare the monthly average engagement rate of tweets containing media with those that did not include media.

Result

Tweets containing **media** consistently achieved higher engagement rates than tweets without media. The highest engagement for media tweets was observed around **June**, while engagement for tweets without media remained comparatively lower throughout the analysed months.

Task 6: Top 10 Tweets by Engagement

Developed a visualization highlighting the Top 10 tweets based on the combined total of Likes and Retweets.

Business Rules Applied

- Weekend tweets excluded.
- Even-numbered impressions.
- Odd-numbered dates.
- Tweet word count below 30.
- Dashboard visible only between 3:00 PM and 5:00 PM IST.

Result

The visualization ranks the **Top 10 tweets** according to total engagement. The highest-performing tweet achieved approximately **90 total engagements**, while the remaining tweets were ranked in descending order. This dashboard helps identify the most engaging content based on user interactions.

5. DAX Measures and Calculated Columns

During the project, several DAX measures and calculated columns were created, including:

- Chart Visibility Measure
- Total Engagement Calculation
- Tweet Category Classification
- Tweet Word Count
- Character Count
- Month and Weekday Sorting Columns
- Quarter Identification
- Media Classification
- Even/Odd Date and Impression Filters
- Time-Based Filtering
- Exclusion Filters for Tweets containing "H" and "D"

6. Skills Developed

Technical Skills

- Microsoft Power BI
- Data Cleaning and Transformation
- Dashboard Development
- Data Visualization
- DAX Functions
- Calculated Columns
- Filters and Slicers

Analytical Skills

- Data Analysis
- Trend Identification
- Critical Thinking
- Problem Solving

Professional Skills

- Technical Documentation
- Report Preparation
- Communication Skills
- Project Execution

7. Key Achievements

- Successfully created six interactive Power BI dashboards.
- Implemented complex business rules using DAX.
- Developed multiple calculated columns and measures.
- Applied dynamic time-based visibility controls.
- Customized sorting for months and weekdays.
- Recorded data limitations while maintaining business requirements.

8. Challenges Encountered and Solutions

- **Challenge:** Implementing chart visibility based on time.
- **Solution:** Used DAX measures to dynamically control visual display.
- **Challenge:** Incorrect chronological order of months and weekdays.
- **Solution:** Created custom sorting columns for accurate sequencing.
- **Challenge:** Managing multiple business conditions simultaneously.
- **Solution:** Applied each filter individually before integrating them into the final report.
- **Challenge:** Empty visualization in Task 1.
- **Solution:** Kept the required conditions unchanged and documented the absence of matching records.

9. Outcomes

The internship significantly enhanced my practical understanding of Power BI, DAX, business intelligence, and dashboard development. Working with real-world Twitter data improved my ability to interpret business requirements, perform data analysis, and present meaningful insights through interactive visualizations.

10. Conclusion

This internship provided valuable hands-on experience in Microsoft Power BI and Business Intelligence. By developing six Twitter analytics dashboards, I gained practical knowledge in data transformation, visualization, DAX calculations, and report creation.

The project also highlighted the importance of adhering to business rules, validating datasets, and documenting observations accurately. Overall, the internship strengthened my technical, analytical, and professional competencies, preparing me for future opportunities in the fields of Data Analytics and Business Intelligence.